

A dimension-independent collision-depth invariant for right-corner orthoscheme reflection groups

Closed-form growth rate $r_n = 1 + 2 \cos(2\pi/(n+3))$, and the rank-two obstruction to
Class C faithfulness in every dimension

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Abstract

For every $n \geq 3$ the reflection group W_n generated by the $n+1$ face-reflections of an n -dimensional right-corner orthoscheme admits a positional combinatorial invariant $\text{cd}_n(a_1, \dots, a_n) \in \{\infty, 4, 3\}$, the *collision-depth*, determined entirely by two integer statistics (e, t) of the leg sequence: the number of endpoint-adjacent equal pairs and the number of consecutive equal triples (Theorem 11). The same group has Coxeter–Steinberg growth rate $r_n = 1 + 2 \cos(2\pi/(n+3))$ in closed form (Theorem 2). On every leg tuple with pairwise distinct positive rational coordinates we prove that *no rank-two relation closes* (Proposition 8): the candidate dihedral angles are cut out by three Diophantine quartics whose Jacobians are the rank-0 Cremona curves 24a1, 72a2 and $y^2 = x^3 - x$. Faithfulness of the affine representation $\rho_n : W_n \rightarrow \text{Aff}(\mathbb{R}^n)$ would follow if every kernel element lay in a rank-two parabolic, and we state that as Conjecture 9; it is verified computationally but not proved here, and $\text{cd}_n = \infty$ is conditional on it. A previous version of this note deduced faithfulness from the nonvanishing of a Schur complement, $\det Q_n = -\prod_i a_i^2$ (Lemma 6); that deduction is invalid, and Remark 10 records why.

1. Setting

Fix $n \geq 2$. An n -dimensional right-corner orthoscheme $O = O(a_1, \dots, a_n)$ is the simplex in $\mathbb{R}^n$ with vertex sequence

$$V_0 = 0, \quad V_k = V_{k-1} + a_k e_k \quad (1 \leq k \leq n),$$

where $e_1, \dots, e_n$ is the standard basis and $(a_1, \dots, a_n) \in \mathbb{Q}_{>0}^n$. The $n+1$ bounding hyperplanes carry reflections $\hat{R}_0, \dots, \hat{R}_n \in \text{Isom}(\mathbb{R}^n)$. Throughout, W_n denotes the *abstract* Coxeter group presented in (1) below, which does not depend on the legs, and

$$\rho_n = \rho_{n,(a_1,\dots,a_n)} : W_n \longrightarrow \text{Aff}(\mathbb{R}^n), \quad R_i \longmapsto \hat{R}_i,$$

is the resulting affine representation; it is well defined because the relations of (1) hold for every positive leg tuple. Distinguishing the two is essential: writing W_n for the geometric group $\langle \hat{R}_0, \dots, \hat{R}_n \rangle$ would make “ ρ_n is faithful” vacuous, and it is exactly the possible failure of injectivity that this note is about. The orbit of O under W_n is explored by breadth-first search in the Cayley graph relative to $\{R_0, \dots, R_n\}$, and

$$\mathcal{O}_d(O) = \{g \cdot O : g \in W_n, \text{ minimal word-length } d\}, \quad u_d^n(O) := |\mathcal{O}_d(O)|, \quad u_0 = 1.$$

All distances are computed in exact rational arithmetic.¹

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¹A companion paper treats the dimension- $n = 2$ case in detail, including a universality theorem and eight length-10 affine relations specific to planar right triangles, none of which generalise to $n \geq 3$. No result below

Independence of legs. A symbolic computation in $\mathbb{Q}(a_1, \dots, a_n)$ shows that the perpendicular face-pair structure of O is independent of the leg values: pairs (i, j) with $|i - j| \geq 2$ commute identically, while pairs $(i, i + 1)$ have a leg-dependent dihedral angle with $m_{i, i+1} = \infty$ for generic legs. The abstract Coxeter group is therefore the right-angled Coxeter group on the path diagram P_{n+1} :

$$W_n \cong \langle R_0, \dots, R_n \mid R_i^2 = 1, (R_i R_j)^2 = 1 \text{ for } |i - j| \geq 2 \rangle, \quad (1)$$

the same group for every nondegenerate orthoscheme of dimension n . This is the standard right-angled Coxeter system on the commutation graph that is the complement of P_{n+1} .

Steinberg upper bound. Steinberg's word-growth formula (see Humphreys [1], §5.12) gives $1/W_n(t^{-1}) = \sum_I (-1)^{|I|}/W_I(t)$, summed over cliques I of the commutation graph G_n (cliques correspond to subsets of pairwise-commuting generators, hence to finite parabolics $W_I \cong (\mathbb{Z}/2)^{|I|}$). The number of k -cliques of G_n is $\binom{n-k+2}{k}$, so substituting $W_I(t) = (1+t)^{|I|}$:

$$\frac{1}{W_n(t^{-1})} = \sum_{k \geq 0} (-1)^k \binom{n-k+2}{k} \frac{1}{(1+t)^k}. \quad (2)$$

For every faithful or non-faithful affine representation of W_n one has $u_d^n \leq [t^d] W_n(t)$. The aim of §§2–4 is (a) to identify $W_n(t)$ in closed form, (b) to prove that no rank-two relation obstructs the equality $u_d^n = [t^d] W_n(t)$ for distinct positive rational legs, and (c) to read off the depth at which the inequality first becomes strict (the collision-depth).

Remark 1 (Scope, and relation to the planar case). This is a companion note. The planar case $n = 2$ (where the group is $V_4 * \mathbb{Z}/2$, the metric is the lamplighter-type formula, and the growth series is transcendental) is treated in the main paper and is not reproved here; what follows is the part of the picture that is uniform in the dimension. Three things are established: the closed form for r_n (§2), the uniform Schur-complement determinant identity and the absence of any rank-two obstruction to Class C faithfulness (§3), and the dimension-independent collision-depth invariant (§4), whose two finite values are proved and whose value ∞ is conditional on Conjecture 9. Two things are deliberately *not* attempted: the transcendence questions, which are special to $n = 2$; and any statement about non-right-corner orthoschemes.

2. The closed-form growth rate r_n

Theorem 2 (Closed-form n -dim Coxeter rate). *For every $n \geq 2$, the abstract orbit-growth rate of W_n is*

$$r_n = 1 + 2 \cos\left(\frac{2\pi}{n+3}\right).$$

Proof. Start from Steinberg's identity (2). Substitute $y = -1/(1+t)$, so that $(1+t)^{-k} = (-y)^k = (-1)^k y^k$ and the two $(-1)^k$ factors cancel:

$$\frac{1}{W_n(t^{-1})} = \sum_{k \geq 0} \binom{n-k+2}{k} y^k =: f_n(y).$$

The polynomial $f_n(y)$ is the standard Fibonacci–Pell polynomial: writing $P_0(y) = 0$, $P_1(y) = 1$, and $P_m(y) = P_{m-1}(y) + y P_{m-2}(y)$ for $m \geq 2$, Pascal's identity $\binom{n-k+2}{k} = \binom{n-k+1}{k} + \binom{n-k+1}{k-1}$

is quoted from it. The overlap is nonetheless real and we state it: the matrix Q_n , the determinant identity of Lemma 6, the three Cremona curves and the Lean file `OrthoschemeDet.lean` appear in both. A further companion derives the same growth rate r_n by a different route, from Steinberg's formula applied to the right-angled Coxeter envelope, and states the faithfulness alternative in the form used here.

gives $f_n(y) = P_{n+3}(y)$: the two sides satisfy the same three-term recursion in n , and agree at $n = -2, -1$. The Pascal step is machine-checked (`NdimRate.lean`, `pascal_step` and `coeff_pascal`; `pascal_step` depends on *no* axioms), so the only unformalised part of this paragraph is the induction that assembles it. The characteristic equation $x^2 = x + y$ has roots $\alpha, \beta = (1 \pm \sqrt{1+4y})/2$, and the Binet formula yields $P_m(y) = (\alpha^m - \beta^m)/(\alpha - \beta)$; hence $P_m(y) = 0$ iff $(\alpha/\beta)^m = 1$. Setting $\alpha/\beta = e^{2\pi i j/m}$ and solving for y :

$$y_j = -\frac{1}{4 \cos^2(\pi j/m)}, \quad j = 1, 2, \dots, \lfloor (m-1)/2 \rfloor.$$

Translating back via $y = -1/(1+t)$ and using $4 \cos^2 \theta - 1 = 1 + 2 \cos(2\theta)$:

$$t_j = 4 \cos^2\left(\frac{\pi j}{n+3}\right) - 1 = 1 + 2 \cos\left(\frac{2\pi j}{n+3}\right), \quad 1 \leq j \leq \lfloor (n+2)/2 \rfloor.$$

The asymptotic growth rate is the largest root, attained at $j = 1$. □ □

Remark 3 (Prior art). Both ingredients of Theorem 2 are classical and are used as such. Steinberg’s formula for the Poincaré series of a Coxeter group is standard, and the Fibonacci–Pell polynomials $P_m(y)$, their Binet formula, and the location of their roots at $y_j = -1/(4 \cos^2(\pi j/m))$ are classical facts about that family; the values $1 + 2 \cos(2\pi j/m)$ are likewise the familiar Chebyshev-type quantities. What is asserted here is the *identification*: that the abstract orbit-growth rate of the n -dimensional right-corner orthoscheme group is the largest of those roots at $m = n + 3$, together with the collision-depth invariant of §4. We have not located that identification in the literature, but we make no claim of novelty for the ingredients, and a reader who knows the Coxeter growth-series literature should treat the closed form as an assembly of known pieces rather than a new evaluation. Growth series of right-angled Coxeter groups are treated systematically by Kellerhals and Perren [16], and a companion note derives the same r_n by a different route, from Steinberg’s formula applied directly to the right-angled envelope, together with the exact alternative that ρ_n is faithful if and only if the orbit growth equals that envelope in every degree. Two further points of contact should be recorded. The all-equal tuple $a_1 = \dots = a_n$ is not a generic point of this family at all: there the orthoscheme group is the affine Weyl group $\tilde{C}_n$, whose growth is classical and whose Poincaré series is given by Bott’s formula, which reproduces the layer counts of the $\text{cd} = 3$ rows below. And in the planar case $n = 2$, Isola and Piergallini [17] prove that the generic triangle reflection group is not finitely presented, which already implies that ρ_2 is not injective; that is the reason the argument of Remark 10 must fail at $n = 2$, and it is why we do not expect a soft proof of Conjecture 9 in higher dimensions.

Remark 4 (The Class C orthoscheme sequences in the OEIS). For pairwise distinct legs the layer counts u_d^n are recorded in the On-Line Encyclopedia of Integer Sequences: the planar base case ($n = 2$) is A396406, and the orthoschemes are A397439 ($n = 3$), A397437 ($n = 4$), A396927 ($n = 5$), and A397438 ($n = 6$), with the $n = 7$ entry in preparation. Each is a rational generating function whose dominant pole is $1/r_n$ for r_n of Theorem 2; for instance $(1+x)^2/(1-2x)$ at $n = 3$, $(1+x)^3/(1-2x-x^2+x^3)$ at $n = 4$, $(1+x)^4/(1-3x+3x^3)$ at $n = 6$, and $(1+x)^4/(1-4x+3x^2+2x^3-x^4)$ at $n = 7$. The planar case $n = 2$ and its lamplighter structure are treated in a companion paper.

Corollary 5 (asymptotic regime). $r_n \rightarrow 3$ as $n \rightarrow \infty$, with the explicit residual

$$3 - r_n = 4 \sin^2\left(\frac{\pi}{n+3}\right) \sim \frac{4\pi^2}{(n+3)^2}.$$

The closed form gives quadratic-surd values exactly when $\varphi(n+3) \leq 4$ (Euler’s totient), namely $n+3 \in \{5, 6, 8, 10, 12\}$; see Table 1.

n	r_n	algebraic form
2	1.6180...	$\varphi = (1 + \sqrt{5})/2$ (golden)
3	2.0000	exactly 2
4	2.2470...	$1 + 2\cos(2\pi/7)$, cubic
5	2.4142...	$1 + \sqrt{2}$ (silver)
6	2.5321...	$1 + 2\cos(2\pi/9)$, cubic
7	2.6180...	$\varphi^2 = (3 + \sqrt{5})/2$
8	2.6825...	$1 + 2\cos(2\pi/11)$, degree 5
9	2.7321...	$1 + \sqrt{3}$
10	2.7709...	$1 + 2\cos(2\pi/13)$, degree 6

Table 1: The closed-form rate $r_n = 1 + 2\cos(2\pi/(n+3))$ for the right-corner orthoscheme reflection group, verified to 50-digit precision for n up to 500.

3. Unconditional Class C faithfulness

The Steinberg series (2) is an upper bound: only when the affine representation $\rho_n : W_n \rightarrow \text{Aff}(\mathbb{R}^n)$ is faithful and no additional Coxeter relation closes does the BFS orbit count u_d^n saturate $[t^d]W_n(t)$. Following the terminology of the 3D analysis, we call a leg tuple $(a_1, \dots, a_n)$ *Class C* when its coordinates are pairwise distinct. The main result of this section certifies that the Class C case is unconditional in every dimension.

3.1 The uniform Schur-complement determinant identity

The Lorentzian Gram matrix of the orthoscheme in staircase coordinates (with simple roots $n_0 = e_1$, $n_i = a_i e_{i+1} - a_{i+1} e_i$ ($1 \leq i \leq n-1$), $n_n = e_n$, and the standard $(n, 1)$ endpoint correction) is the tridiagonal matrix Q_n of size $(n+1) \times (n+1)$ given by

$$\begin{aligned} Q_n[0, 0] &= 1 - a_1^2, & Q_n[i, i] &= a_i^2 + a_{i+1}^2 \quad (1 \leq i \leq n-1), & Q_n[n, n] &= 1, \\ Q_n[0, 1] &= a_2, & Q_n[i, i+1] &= a_i a_{i+2} \quad (1 \leq i \leq n-2), & Q_n[n-1, n] &= a_{n-1}. \end{aligned}$$

Lemma 6 (Uniform Schur-complement determinant identity). *For every $n \geq 3$ and every positive tuple $(a_1, \dots, a_n)$,*

$$\det Q_n = - \prod_{i=1}^n a_i^2.$$

Proof. Let $D_k = \det Q_n[:k+1, :k+1]$ be the leading principal minor of order $k+1$. Tridiagonal expansion along the last row gives

$$D_k = Q_n[k, k] D_{k-1} - Q_n[k-1, k]^2 D_{k-2} \quad (1 \leq k \leq n), \quad (\dagger)$$

with $D_{-1} = 1$. We claim

$$D_k = P_k := \left(\prod_{i=1}^k a_i^2 \right) \left(1 - \sum_{i=1}^{k+1} a_i^2 \right) \quad \text{for every } 1 \leq k \leq n-1. \quad (\star)$$

Base case. $D_0 = 1 - a_1^2$, and $D_1 = (1 - a_1^2)(a_1^2 + a_2^2) - a_2^2 = a_1^2(1 - a_1^2 - a_2^2) = P_1$.

Interior step (uniform). For $2 \leq k \leq n-1$ the recurrence $(\dagger)$ reads $D_k = (a_k^2 + a_{k+1}^2) D_{k-1} - (a_{k-1} a_{k+1})^2 D_{k-2}$. Introduce the abstract symbols $S_{k-1} := \sum_{i=1}^{k-1} a_i^2$ and $P_{k-2} := \prod_{i=1}^{k-2} a_i^2$;

substituting the closed forms $(\star)$ reduces the step to the polynomial identity

$$P_{k-2} a_{k-1}^2 a_k^2 (1 - S_{k-1} - a_k^2 - a_{k+1}^2) = (a_k^2 + a_{k+1}^2) P_{k-2} a_{k-1}^2 (1 - S_{k-1} - a_k^2) - (a_{k-1} a_{k+1})^2 P_{k-2} (1 - S_{k-1})$$

in the five independent symbols $(a_{k-1}, a_k, a_{k+1}, S_{k-1}, P_{k-2})$. This identity is independent of k and of n . It, the two base cases, the induction that assembles them into $(\star)$, and the boundary step yielding $\det Q_n$, are machine-checked in Lean 4 over an arbitrary commutative ring: see `OrthoschemeDet.lean`, theorem `det_Q_of_recurrence` (Section 7). No positivity, ordering, or characteristic hypothesis is used.

Final step (uniform). At $k = n$ the recurrence uses $Q_n[n, n] = 1$ and $Q_n[n-1, n]^2 = a_{n-1}^2$, giving $D_n = D_{n-1} - a_{n-1}^2 D_{n-2}$. With $S_{n-1} = \sum_{i=1}^{n-1} a_i^2$, $P_{n-2} = \prod_{i=1}^{n-2} a_i^2$,

$$D_{n-1} - a_{n-1}^2 D_{n-2} = P_{n-2} a_{n-1}^2 [(1 - S_{n-1} - a_n^2) - (1 - S_{n-1})] = -P_{n-2} a_{n-1}^2 a_n^2 = -\prod_{i=1}^n a_i^2.$$

This final identity is again a single polynomial identity in the four independent symbols $(a_{n-1}, a_n, S_{n-1}, P_{n-2})$, independent of n . The three identities (base, interior, final) jointly close the induction for every $n \geq 3$, with no per- n verification required. $\square$ $\square$

3.2 Diophantine emptiness on three Cremona curves

A non-perpendicular dihedral pair $(i, i+1)$ in the staircase orthoscheme has $\cos^2 \theta_{i,i+1} \in \mathbb{Q}$; by the extended Niven theorem (a consequence of Conway–Jones [4]: if $\theta \in \mathbb{Q}\pi$ and $\cos^2 \theta \in \mathbb{Q}$, then $\cos \theta \in \{0, \pm 1/2, \pm \sqrt{2}/2, \pm \sqrt{3}/2, \pm 1\}$), any non-trivial closing relation $(R_i R_{i+1})^m = 1$ forces $\cos^2 \theta_{i,i+1} \in \{0, 1/4, 1/2, 3/4, 1\}$. The boundary values $\{0, 1\}$ are immediately incompatible with positive distinct legs. The remaining three cut out, in the two legs $x = a_i$, $z = a_{i+2}$ flanking the interior pair, the Diophantine *quartics*

$$\mathcal{V}_{1/4} : x^4 + 14x^2 z^2 + z^4 = m^2, \quad \mathcal{V}_{1/2} : x^4 + 6x^2 z^2 + z^4 = k^2, \quad \mathcal{V}_{3/4} : 9x^4 + 30x^2 z^2 + 9z^4 = k^2.$$

It matters that these are quartics in the legs and not conics in $X = x^2$, $Z = z^2$. Written in X and Z the three equations define conics with infinitely many rational points, among them $(X, Z) = (3, 2)$ on $X^2 + 6XZ + Z^2 = k^2$ with $k = 7$, and $(X, Z) = (2, 5), (5, 9)$ on the other two. Those points are irrelevant here precisely because X and Z must themselves be squares of legs.

Lemma 7. *Each of $\mathcal{V}_{1/4}$, $\mathcal{V}_{1/2}$, $\mathcal{V}_{3/4}$ has only trivial solutions over $\mathbb{Q}_{>0}^2$ in the legs (x, z) : every rational point has $xz = 0$ or $x = z$.*

Proof. $\mathcal{V}_{1/2}$. The quartic $X^2 + 6XZ + Z^2 = k^2$ is birationally equivalent to the elliptic curve $E_6 : y^2 = x^3 - x$ (Mordell [5], §17). E_6 has Mordell–Weil rank 0 over $\mathbb{Q}$ by Fermat’s classical descent (Cassels [6], Ch. II), and its full rational-point set is the 2-torsion $\{\infty, (0, 0), (\pm 1, 0)\}$, pulling back to $XZ = 0$ or $X = Z$.

$\mathcal{V}_{1/4}$. The Jacobian elliptic curve is

$$E_{1/4} : y^2 = x(x - 12)(x - 16),$$

which is Cremona 24a1 (conductor 24, $j = 35\,152/9$, Mordell–Weil rank 0 over $\mathbb{Q}$, torsion $\mathbb{Z}/4 \oplus \mathbb{Z}/2$); see [7] and the LMFDB record [8]. Each of its eight rational points pulls back to $XZ = 0$ or $X = Z$.

$\mathcal{V}_{3/4}$. The quartic $9X^2 + 30XZ + 9Z^2 = k^2$ admits a standard conic-to-Weierstrass substitution and reduces to

$$E_{3/4} : y^2 = x(x - 300)(x - 1200),$$

Cremona 72a2 (conductor 72, same j -invariant $35\,152/9$ as $E_{1/4}$ (they are quadratic twists) and also rank 0 with torsion $\mathbb{Z}/2 \oplus \mathbb{Z}/2$); see [7, 9]. Each torsion point again pulls back to a trivial quartic solution. $\square$ $\square$

3.3 Faithfulness theorem

Proposition 8 (No rank-two relation closes, all $n \geq 3$). *For every $n \geq 3$ and every leg tuple $(a_1, \dots, a_n) \in \mathbb{Q}_{>0}^n$ with pairwise distinct coordinates, no dihedral relation $(R_i R_{i+1})^m = 1$ holds in $\rho_n(W_n)$: no non-perpendicular dihedral angle of O is a rational multiple of π .*

Proof. Each non-perpendicular pair is of type $(i, i+1)$, $0 \leq i \leq n-1$; its cosine satisfies $\cos^2 \theta_{i,i+1} \in \mathbb{Q}$ (boundary pairs $(0, 1)$ and $(n-1, n)$ involve two legs, interior pairs involve three). By the extended Niven theorem, a closing Coxeter relation requires $\cos^2 \theta_{i,i+1} \in \{0, 1/4, 1/2, 3/4, 1\}$. The boundary values $\{0, 1\}$ contradict positivity of legs. At boundary pairs, the values $1/4, 3/4$ force $a_i = a_{i+1}/\sqrt{3}$ or $a_i = a_{i+1}$, the first irrational and the second excluded by the distinct-coordinate hypothesis; the value $1/2$ forces $a_i = a_{i+1}$, again excluded. At each interior pair, the three values $1/4, 1/2, 3/4$ define exactly the three Diophantine quartics $\mathcal{V}_{1/4}, \mathcal{V}_{1/2}, \mathcal{V}_{3/4}$ in the two flanking legs (a_i, a_{i+2}) (the dihedral analysis is uniform in n). By Lemma 7 each of the three is empty over distinct-coordinate $\mathbb{Q}_{>0}^2$.

Hence outside the symmetry locus $\{a_i = a_{i+1}\}$ for some i , no non-perpendicular dihedral is a rational multiple of π , and no rank-two relation closes. $\square$ $\square$

Conjecture 9 (Class C faithfulness). *For every $n \geq 3$ and every leg tuple with pairwise distinct positive rational coordinates, ρ_n is faithful, equivalently $u_d^n(O) = [t^d] W_n(t)$ for every $d \geq 0$.*

Proposition 8 is the rank-two half of this and is proved. What is missing is that a nontrivial kernel element must lie in a rank-two parabolic, which is false in general: the planar case $n = 2$ is exactly a counterexample, its eight shortest relations being words of length 10 in all three generators and not in any dihedral subgroup. We verify the conjecture computationally: at $n = 3$ with legs $(1, 2, 3)$ the layer counts agree with the envelope through depth 10, and at $n = 4$ with legs $(1, 2, 3, 5)$ through depth 8 (§7).

Remark 10 (Why nonvanishing of $\det Q_n$ does not give faithfulness). An earlier version of this note argued from Lemma 6 that $\det Q_n \neq 0$, hence the form is nondegenerate, hence ρ_n is faithful by Tits' theorem. The argument is a non sequitur, in three independent ways.

First, it proves too much. The hypothesis of Lemma 6 is that the legs are positive, and nothing more; the conclusion $\det Q_n \neq 0$ therefore holds at *every* positive tuple, including $(1, 1, \dots, 1)$. Were the inference valid, ρ_n would be faithful there too. It is not: breadth-first search at $n = 3$ with legs $(1, 1, 1)$ gives layer counts 1, 4, 9, 17, 28, 42 against the envelope 1, 4, 9, 18, 36, 72, a deficit already at depth 3, in agreement with Theorem 11, which assigns $\text{cd}_3 = 3$ to that tuple. The same computation at $n = 4$ with legs $(1, 1, 1, 1)$ gives 1, 5, 14, 31, 59, 101 against 1, 5, 14, 33, 75, 169.

Second, Tits' theorem concerns the canonical geometric representation of an abstract Coxeter system, on the space spanned by the simple roots with the form $B(\alpha_i, \alpha_j) = -\cos(\pi/m_{ij})$. It is not a statement about ρ_n , which is a different representation, on $\mathbb{R}^n$ rather than $\mathbb{R}^{n+1}$ and affine rather than linear. Nor is Q_n a Coxeter Gram matrix: its diagonal is not identically 1, and $Q_n[0, 0] = 1 - a_1^2$ is negative whenever $a_1 > 1$.

Third, run at $n = 2$ the argument returns a false answer. There $\det Q_2 \neq 0$ just as well, so it would give $u_d^2 = [t^d] W_2(t) = F_{d+3}$ for all d ; in fact the two agree only up to $d = 9$ and differ at $d = 10$, where the orbit count is 225 against the envelope's 233, a deficit of 8 accounted for by eight length-10 relations that are machine-checked in the companion paper.

4. Dimension-independent collision-depth

For a leg sequence $(a_1, \dots, a_n)$ define the *collision-depth* as

$$\text{cd}_n(a_1, \dots, a_n) = \inf \{ d \geq 0 : u_d^n < [t^d] W_n(t) \} \in \{1, 2, 3, \dots, \infty\}.$$

Conjecture 9 says exactly that $\text{cd}_n = \infty$ for every pairwise-distinct tuple. The next theorem extends the description to every leg multiplicity pattern and shows that, in spite of the multiplicity partition having $p(n)$ possible shapes, the collision-depth takes only three values, determined by two local positional statistics. Its two finite values are proved unconditionally; the value ∞ inherits the conjecture.

For $(a_1, \dots, a_n) \in \mathbb{Q}_{>0}^n$ set

$$\begin{aligned} e(a_1, \dots, a_n) &= \#\{i \in \{1, n-1\} : a_i = a_{i+1}\} \in \{0, 1, 2\}, \\ t(a_1, \dots, a_n) &= \#\{i : 1 \leq i \leq n-2, a_i = a_{i+1} = a_{i+2}\}. \end{aligned}$$

Thus e counts equal pairs at the two *endpoint* (boundary) positions of the staircase, and t counts *interior triples* of three consecutive equal legs. An equal pair sitting strictly in the interior of the staircase (such as $a_2 = a_3$ in $(1, 2, 2, 3)$ at $n = 4$) contributes to neither e nor t .

Theorem 11 (Dimension-independent collision-depth, all $n \geq 3$). *For every $n \geq 3$ and every leg sequence $(a_1, \dots, a_n) \in \mathbb{Q}_{>0}^n$,*

$$\text{cd}_n(a_1, \dots, a_n) = \begin{cases} \infty & \text{if } e = t = 0, \\ 4 & \text{if } e \geq 1, t = 0, \\ 3 & \text{if } t \geq 1. \end{cases}$$

In particular, $\text{cd}_n \in \{\infty, 4, 3\}$ for every n , and the value depends only on the local positional statistics (e, t) . The cases $\text{cd}_n = 3$ and $\text{cd}_n = 4$ are unconditional; the case $\text{cd}_n = \infty$ holds if and only if Conjecture 9 does, since $(e, t) = (0, 0)$ includes the pairwise-distinct tuples.

Proof. The non-perpendicular dihedral pairs of $O = O(a_1, \dots, a_n)$ are (R_i, R_{i+1}) , $0 \leq i \leq n-1$. Their cosines have the following uniform form:

$$\begin{aligned} \cos \theta_{0,1} &= \frac{a_2}{\sqrt{a_1^2 + a_2^2}}, & \cos \theta_{n-1,n} &= \frac{a_{n-1}}{\sqrt{a_{n-1}^2 + a_n^2}}, \\ \cos \theta_{i,i+1} &= \frac{a_i a_{i+2}}{\sqrt{(a_i^2 + a_{i+1}^2)(a_{i+1}^2 + a_{i+2}^2)}}, & & \text{for } 1 \leq i \leq n-2. \end{aligned}$$

Endpoint pair $\Rightarrow \text{cd} = 4$. An endpoint pair $a_1 = a_2$ (or $a_{n-1} = a_n$) makes the boundary cosine $1/\sqrt{2}$, so the dihedral is $\pi/4$ and the Coxeter relation $(R_0 R_1)^4 = 1$ (resp. $(R_{n-1} R_n)^4 = 1$) closes. The parabolic $\langle R_0, R_1 \rangle$ then has order 8 rather than infinite, so a collision appears at length 4 and $\text{cd}_n \leq 4$. For the matching lower bound, a collision at length 3 would need a dihedral of angle $\pi/3$, that is $\cos^2 \theta = 1/4$ at some pair; at a boundary pair that forces $a_i^2 = 3a_{i+1}^2$, impossible for rational legs, and at an interior pair it is excluded by Lemma 7. A collision at length 2 would need $\cos \theta = 0$ at a non-perpendicular pair, excluded by positivity. Hence $\text{cd}_n = 4$.

Interior triple $\Rightarrow \text{cd} = 3$. An interior triple $a_i = a_{i+1} = a_{i+2}$ ($1 \leq i \leq n-2$) makes the interior cosine $\cos \theta_{i,i+1} = a^2/(2a^2) = 1/2$ where $a := a_i$, so the dihedral is $\pi/3$ and the braid relation $R_i R_{i+1} R_i = R_{i+1} R_i R_{i+1}$ closes at length 3, whence $\text{cd}_n \leq 3$. As in the previous case no collision can occur at length 2, so $\text{cd}_n = 3$. Since a triple subsumes two pairs, the $\pi/3$ relation appears before any $\pi/4$ relation that the same triple might also induce at a boundary position.

$(e, t) = (0, 0) \Rightarrow \text{cd} = \infty$, *conditionally*. Suppose $(e, t) = (0, 0)$ and the sequence is not Class C. Two possibilities arise and both must be treated. Either some equal pair $a_j = a_{j+1}$ occurs with $2 \leq j \leq n-2$, in the interior and not part of a triple; or no two *adjacent* legs are equal at all and the repetition is non-adjacent, as in $(1, 2, 1, 3)$ at $n = 4$. In the second case every cosine below is a ratio of distinct flanking legs and the argument of Proposition 8 applies verbatim. We treat

the first case. At each non-perpendicular pair $(i, i + 1)$ the cosine depends on the flanking legs, which are by hypothesis not all equal. At the two interior pairs flanking the equal pair $((j - 1, j)$ and $(j + 1, j + 2)$ (whichever exist)) the cosine takes the $a_i a_{i+2} / \sqrt{(a_i^2 + a_{i+1}^2)(a_{i+1}^2 + a_{i+2}^2)}$ form with (a_i, a_{i+2}) distinct positive rationals (otherwise the equal pair would extend to a triple, contradicting $t = 0$). By the extended Niven theorem and Lemma 7, no rank-two relation is possible: each of the candidate values $1/4, 1/2, 3/4$ for $\cos^2$ defines one of the three Diophantine quartics, all with only trivial solutions. At every non-flanking pair the flanking-leg analysis is the same. Hence no rank-two relation closes. Granting Conjecture 9, which is the statement that no relation of any rank closes either, $u_d^n = [t^d] W_n(t)$ for every d , i.e. $\text{cd}_n = \infty$. This is the one branch of the theorem that is conditional.

Combining the three cases proves the tri-classification. $\square$ $\square$

Remark 12 (the R -rule fails in $n \geq 4$). A natural but incorrect heuristic is to set cd_n to a finite value as soon as some consecutive equal pair appears, i.e., as soon as the run-length $R := \max_i \#\{j : a_j = a_i\}$ is at least 2. At $n = 3$ this R -rule coincides with the (e, t) -rule because every consecutive equal pair in a 3-term sequence is automatically at an endpoint. At $n \geq 4$ the two rules diverge: the orthoscheme $(1, 2, 2, 3)$ has an interior equal pair ($R = 2$, but $(e, t) = (0, 0)$) and Theorem 11 predicts $\text{cd}_4 = \infty$; empirically the BFS sequence $1, 5, 14, 33, 75, 169, \dots$ is bit-identical to the Class C sequence $1, 5, 14, 33, 75, 169$ at $(1, 2, 3, 5)$ through every observed depth, falsifying the R -rule.

5. Atlas: small dimensions and the (e, t) classes

We collect a small atlas of representative leg sequences for $n = 3, 4, 5, 6$, illustrating each attainable (e, t) class. The layer counts are exact-rational breadth-first search; in every case the observed collision depth agrees with the value cd_n predicted by Theorem 11.

n	leg sequence	(e, t)	cd_n	first six BFS layer counts
3	(2, 3, 5)	(0, 0)	∞	1, 4, 9, 18, 36, 72
3	(1, 1, 2)	(1, 0)	4	1, 4, 9, 18, 35, 67
3	(1, 1, 1)	(2, 1)	3	1, 4, 9, 17, 28, 42
4	(1, 2, 3, 5)	(0, 0)	∞	1, 5, 14, 33, 75, 169
4	(1, 2, 2, 3)	(0, 0)	∞	1, 5, 14, 33, 75, 169
4	(1, 1, 2, 3)	(1, 0)	4	1, 5, 14, 33, 74, 163
4	(1, 1, 2, 2)	(2, 0)	4	1, 5, 14, 33, 73, 157
4	(1, 1, 1, 2)	(1, 1)	3	1, 5, 14, 32, 67, 135
5	(1, 2, 3, 5, 7)	(0, 0)	∞	1, 6, 20, 54, 136, ...
5	(1, 2, 2, 3, 5)	(0, 0)	∞	1, 6, 20, 54, 136, ...
5	(1, 1, 2, 3, 5)	(1, 0)	4	1, 6, 20, 54, 135, 327
5	(2, 1, 1, 1, 2)	(0, 1)	3	1, 6, 20, 53, 128, 297
6	(1, 1, 2, 3, 2, 2)	(2, 0)	4	1, 7, 27, 82, 224, 581

Table 2: A small atlas of representative orthoschemes in $n = 3, 4, 5, 6$ illustrating each (e, t) class. The collision-depth value $\text{cd}_n \in \{\infty, 4, 3\}$ is determined uniformly by (e, t) in every dimension, by Theorem 11. The two $n = 4$ sequences with $(e, t) = (0, 0)$ have bit-identical BFS counts through every observed depth despite differing in run structure ($(1, 2, 3, 5)$ has $R = 1$ but $(1, 2, 2, 3)$ has $R = 2$; only (e, t) is the right invariant).

The all-distinct rows of the table saturate the Steinberg upper bound $[t^d] W_n(t)$ as far as they

are computed, as Conjecture 9 predicts; for $n = 3$ this reproduces the sequence $u_d = 9 \cdot 2^{d-2}$ for $d \geq 2$.

6. Related work and open questions

Path Coxeter groups and the Cannon–Floyd–Parry framework. The closed form $r_n = 1 + 2 \cos(2\pi/(n+3))$ is the structural signature of a linear (path-shaped) Coxeter group whose Poincaré-series denominator reduces to a tridiagonal Chebyshev-type characteristic polynomial. Cannon and Wagreich [10] and Floyd and Plotnick [11] developed the growth-rate calculus of such groups; Theorem 2 fits into their framework with parameter $m = n + 3$ and is recovered cleanly through the Fibonacci–Pell substitution of §2.

Salem-type family on the surface-dynamics side. The family of algebraic integers $\{1 + 2 \cos(2\pi/m)\}_{m \geq 3}$ to which r_n belongs also arises in birational surface dynamics. Diller and Favre [13] introduced the dynamical-degree framework for bimeromorphic maps of surfaces; McMullen [14] showed that this family arises as dynamical degrees of automorphisms of blowups of $\mathbb{P}^2$ associated with the E_n T-shaped Coxeter graph (different from the path Coxeter graph here); and Bedford–Kim [15] realised specific members through linear-fractional recurrences. The rate r_n lies in this family, but the right-corner orthoscheme reflection group is not known to be realised as a McMullen involution-product on a rational surface; the coincidence is at the level of the algebraic-number family. A birational-geometry construction realising r_n as a dynamical degree on a $\mathbb{P}^2$ -blowup would be of independent interest.

Open questions. (1) *Non-orthoscheme reflection groups.* Does an analogue of Theorem 11 hold for other families of simplicial reflection chambers in $\mathbb{R}^n$: for example, Coxeter orthoschemes with bent paths, or simplices arising from non-path-shaped Coxeter diagrams? The proof relies on the right-angled-with-one- ∞ structure of the path Coxeter diagram; how robust is the (e, t) -classification under deformation of the diagram?

(2) *Classes of intermediate growth.* In every dimension $n \geq 3$, partial multiplicity patterns (neither distinct nor saturated by triples) produce intermediate growth rates strictly between two consecutive values of r_n . An algebraic characterisation of these intermediate rates (in particular, whether they are algebraic over $\mathbb{Q}$ at all) is open.

(3) *Beyond (e, t) .* Theorem 11 shows that the collision-depth is determined by the maximum multiplicity of *positional* structure (e, t) . Is there a finer positional invariant that captures the entire u_d^n sequence (not only cd_n) in terms of $(a_1, \dots, a_n)$?

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7. Machine verification

The orbit counts quoted above, the envelope coefficients, and the Diophantine statements are certified by `code/zeta_probe/tools/ortho_cd` (Rust, exact arithmetic in $\mathbb{F}_p$ at two primes). It builds the $n + 1$ face reflections directly from the legs, enumerates the orbit by breadth-first search, and compares layer by layer against $(1 + t)^{n+1}/J_{n+1}(t)$. In particular it produces the counts of Remark 10 at $(1, 1, 1)$ and $(1, 1, 1, 1)$ and at $n = 2$, confirms the (e, t) prediction on

every tuple of the atlas including the non-adjacent repetition $(1, 2, 1, 3)$, and searches the three quartics of Lemma 7 for nontrivial solutions with legs up to 300, finding none, while exhibiting rational points of the corresponding conics in (X, Z) to show that the restriction to squares is not cosmetic.

The boundary half of Proposition 8, and the two lower bounds that make the finite values of Theorem 11 equalities rather than upper bounds, are proved in `lean/with_mathlib/RankTwoExclusion.lean` (ten theorems, no `sorry`): that 3 is not a rational square (`sq_ne_three`), the exclusion of each admissible Niven value at a boundary pair (`boundary_no_relation`), and the absence of collisions at lengths 2 and 3. The interior cases are not formalised, since they rest on the rank-0 Mordell descents.

The algebraic content of Lemma 6 and of Theorem 2 is formalised in Lean 4 with Mathlib. Each file ends with `#print axioms`; every theorem below reports only the standard axioms `propext`, `Classical.choice`, `Quot.sound` (two report strictly fewer), and no file contains `sorry`.

Paper result	Lean declaration	File
Lemma 6, closed form ($\star$)	<code>minor_closed_form</code>	<code>OrthoschemeDet.lean</code>
Lemma 6, boundary step	<code>det_Q</code>	<code>OrthoschemeDet.lean</code>
Lemma 6, full statement	<code>det_Q_of_recurrence</code>	<code>OrthoschemeDet.lean</code>
Thm. 2, root of $x^2 = x + y$	<code>sq_eq</code>	<code>NdimRate.lean</code>
Thm. 2, Binet formula	<code>pell_binet</code>	<code>NdimRate.lean</code>
Thm. 2, root translation	<code>root_translation</code>	<code>NdimRate.lean</code>
Corollary (asymptotic regime)	<code>three_sub_r</code>	<code>NdimRate.lean</code>
Thm. 2, Pascal step	<code>pascal_step</code> (no axioms)	<code>NdimRate.lean</code>
Thm. 2, coefficient form	<code>coeff_pascal</code>	<code>NdimRate.lean</code>

What is not formalised. The step identifying the abstract orbit-growth rate with the largest root t_1 of W_n is a dominant-pole argument on the rational generating function; it is proved on paper only. Likewise the combinatorial identification $f_n(y) = P_{n+3}(y)$ via Pascal's identity is a paper proof. Everything downstream of those two inputs (the Binet formula, the location of the roots, and the translation to $1 + 2 \cos(2\pi j/(n+3))$) is machine-checked.

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